NORTHWIND TRADERS: TECHNICAL ARCHITECTURE & ANALYST GUIDE

End-to-End Sales Performance & Employee Efficiency Model

This document details the transition of Northwind Traders data from raw CSV to MySQL and finally into a Star Schema-based Power BI dashboard. This document is articulated for Analysts and Analytical Managers.

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Contents

1. Project Overview & Scope
   1.1. Executive Summary
   1.2. Key Project Baselines
2. Data Engineering & ETL
   2.1. Source System & Migration
   2.2. Power Query Transformations
3. Data Modelling Architecture
   3.1. Star Schema Implementation
      3.1.1. Table Roles
   3.2. Relationship Logic
4. DAX Measure Dictionary
   4.1. Revenue Calculation
   4.2. Efficiency Metrics
5. UI/UX & Accessibility Audit
   5.1. Visual Hierarchy
   5.2. Accessibility for Diverse Users
6. AI-Augmented Analytical Methodology
   6.1. The AAR Principle (Aid, Advice, Recommend)
   6.2. The SDD Framework (Support, Discussion, Dependencies)

1. Project Overview & Scope

1.1.Executive Summary

The objective of this project was to engineer a robust Business Intelligence solution for Northwind Traders, transitioning from decentralized raw data to an interactive, multi-page Power BI dashboard.

1.2.Key Project Baselines

- **Total Revenue Analyzed:** $1.27M
- **Transaction Volume:** 830 Total Orders
- **Temporal Scope:** July 2013 to May 2015
- **Primary Metrics:** Average Order Value ($1.53K) and Overall Freight Costs ($78.24).

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2. Data Engineering & ETL

2.1.Source System & Migration

The raw data originated as standard relational CSV files. To ensure data integrity and practice SQL-to-BI connectivity, the data was migrated into a **MySQL database** environment. This allowed for the validation of Primary and Foreign Key relationships before ingestion.

2.2.Power Query Transformations

- **Date Normalization:** The Order Date field was standardized to a "Date" type to support the report's timeline slicers.
- **Measure Isolation:** A dedicated Project Measures table was created to store all manual DAX calculations, ensuring a clean and organized metadata structure for future developers.
- **Locale & Localization**: The Power BI file was configured with the **English (India)** locale. This ensures that the **July 2013 to May 2015** temporal scope is consistently interpreted in the **DD-MM-YYYY** format across all stakeholder regions.

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3. Data Modelling Architecture

3.1.Star Schema Implementation

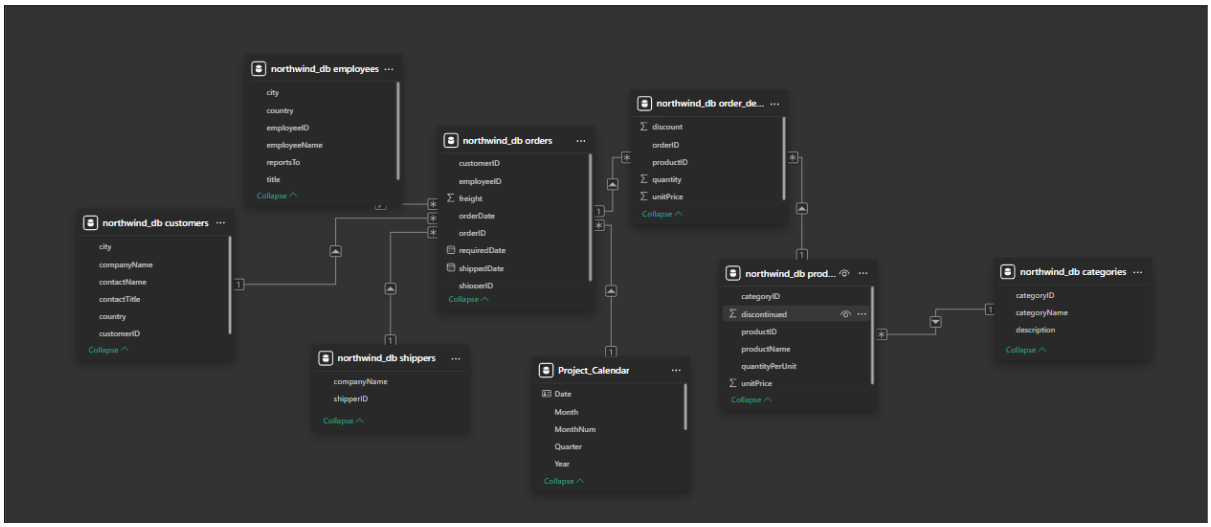

Figure 1 Northwind Traders Star Schema Architecture. This diagram illustrates the relational data model optimized for Power BI performance. It features a central Fact Table (order_details) connected to five primary Dimension Tables via One-to-Many (1:) relationships. This schema ensures efficient filter propagation across the $1.27M in total revenue and prevents data ambiguity during multi-dimensional analysis.

The model follows a high-performance **Star Schema** to optimize query speed and filter accuracy.

3.1.1. Table Roles

- **Fact Table:** order_details (Contains quantity, unit price, and discount).
- **Dimension Tables:** Employees, Customers, Products, Shippers, Orders, and Project_Calendar.
- ▯ **Grain: Line-Item Level** (2,155 rows).

3.2.Relationship Logic

All relationships are configured as **One-to-Many (1:*)** with single-directional cross-filtering. This architecture prevents "Circular Dependencies" and ensures that the **Total Revenue ($1.27M)** remains consistent regardless of the filters applied.

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4. DAX Measure Dictionary

4.1.Revenue Calculation

To account for row-level discounts, a manual iterator was used:

- **Formula:** `Total Revenue = SUMX('northwind_db order_details', 'northwind_db order_details'[unitPrice] * 'northwind_db order_details'[quantity] * (1 - 'northwind_db order_details'[discount]))`

- **Analyst Note:**

  - This ensures the 15%–25% discounts offered to top customers are accurately reflected in the final revenue total.

  - A standard SUM aggregation was bypassed in favour of the SUMX iterator. This architectural choice was necessary to maintain "Row-Level Integrity," ensuring that the **15%-25% discounts** were deducted from each line item before being aggregated into the final **$1.27M revenue** baseline.

4.2.Efficiency Metrics

- **Orders Per Employee:** `DIVIDE([Total Orders], COUNTROWS('northwind_db employees'),0)`

- **Average Order Value:** `DIVIDE([Total Revenue], [Total Orders], 0)`

- **Logic:** The `DIVIDE` function was used to handle "divide by zero" errors gracefully, resulting in a staff benchmark of **92 orders per employee**.

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5. UI/UX & Accessibility Audit

5.1.Visual Hierarchy

Figure 2 Executive Sales Overview. Highlighting the F-Pattern layout. Note the placement of high-level KPIs ($1.27M Revenue) in the top-left for immediate visibility, and the monochromatic blue saturation gradient used in the Customer Treemap to ensure colorblind accessibility.

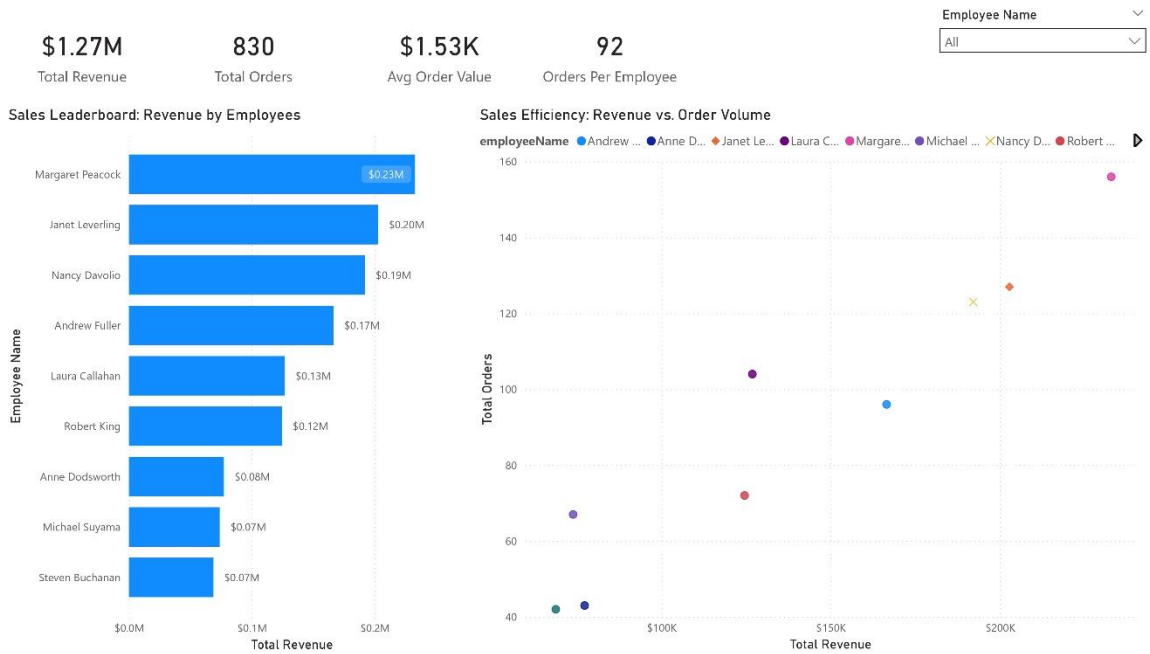

Figure 3 Employee Performance & Efficiency View. This page demonstrates the use of varied marker shapes (Circles, Squares, X-markers) in the Scatter Plot, allowing for data differentiation without relying on color perception. It also showcases the custom DAX calculation for Orders Per Employee (92).

The report utilizes the **F-Pattern** layout, placing critical KPIs in the top-left quadrant for maximum executive visibility.

5.2.Accessibility for Diverse Users

The dashboard was audited for color blindness (Protanopia/Deuteranopia):

- **Treemap Saturation:** The Customer Revenue Treemap uses a single-hue blue gradient.
- **Marker Diversity:** The Sales Efficiency scatter plot uses unique **Shapes (Markers)** for each employee to ensure data is readable in grayscale.

6. AI-Augmented Analytical Methodology

This project utilized Large Language Models (LLMs) as a strategic technical consultant. The core data comprehension, architectural decision-making, and analytical flow remained entirely human-directed, ensuring strict data governance and logical integrity from the raw CSVs to the final Star Schema dashboard.

6.1. The AAR Principle (Aid, Advice, Recommend)

AI was leveraged systematically to accelerate development without compromising technical rigor:

- **Aid (Syntax & Structuring):** AI aided in formatting complex MySQL validation queries and structuring DAX iterators, such as the SUMX revenue logic ($1.27M) and DIVIDE efficiency metrics (92 orders per employee).
- **Advice (Best Practices):** The AI provided industry-standard advice on Power BI architecture, reinforcing the decision to isolate metrics into a dedicated **Project Measures** table and generate a DAX-based **Project_Calendar** dimension for the July 2013 to May 2015 scope.
- **Recommend (UI/UX Strategy):** AI recommendations were factored into the final presentation layer, specifically regarding the **F-Pattern** layout for executive scannability and the colourblind accessibility audits applied to the customer treemap and scatter plots.

6.2. The SDD Framework (Support, Discussion, Dependencies)

To maintain an unbroken and verified analytical flow, AI interactions were governed by the SDD framework:

- **Support (Targeted Debugging):** AI acted as a debugging support tool for specific visual behaviours, such as resolving chronological axis sorting (Q1 → Q4) to maintain temporal

integrity. Every supported fix was manually tested and verified in the model before execution.

- **Discussion (Analytical Narrative):** Before writing code, AI was used as a sounding board to discuss and refine the analytical narrative, ensuring the transition from MySQL to the **Employee Efficiency Model** addressed specific managerial needs.
- **Dependencies (Strict Human Oversight):** The project maintained a "zero-trust" dependency on AI for raw logic. As a Primary Data Analyst, my role was to understand the Northwind dataset, define the execution steps, and validate all outcomes against the underlying data relationships to ensure 100% accuracy.

Appendix A: Technical Glossary

This glossary is provided to assist junior analysts and non-technical stakeholders in understanding the core data engineering and UI/UX concepts utilized in this project.

- **DAX (Data Analysis Expressions):** The formula language used in Power BI to create custom calculations. In this project, iterators like SUMX and DIVIDE were used to handle row-level discounts and prevent division errors.
- **Dimension Table:** A lookup table that contains descriptive attributes used to filter and group data, such as Employees, Customers, and the Project_Calendar.
- **F-Pattern Layout:** A UI/UX design strategy that places critical KPIs (like the **$1.27M Total Revenue**) in the top-left quadrant for immediate visibility.
- **Fact Table:** The central table in a data model containing quantitative, measurable data. In this project, order_details act as the Fact Table.
- **LLM (Large Language Model):** An advanced AI model used as a strategic technical consultant via the **AAR** and **SDD** frameworks.
- **One-to-Many (1:*) Relationship:** A database connection where a single record in a Dimension table relates to multiple records in a Fact table.
- **Star Schema:** A data modelling architecture where a central Fact table is surrounded by Dimension tables to optimize performance and filter accuracy.
- **Grain (Granularity):** The level of detail contained in a single row of a table. The Orders table has a **Coarse Grain** (one row per 830 transactions), while the order_details table

has a **Fine Grain** (one row per 2,155 product line items). Understanding grain is critical for ensuring that **Total Revenue ($1.27M)** is aggregated correctly across different dimensions

Appendix B: Source Code Snippet (SQL Validation)

To ensure the **830 Total Orders** were valid before ingestion, the following SQL logic was used in the MySQL environment to audit the schema:

SQL

```
-- Validating Foreign Key integrity and Revenue Baseline

SELECT

    COUNT(od.orderID) AS total_line_items,

    SUM(od.unitPrice * od.quantity * (1 - od.discount)) AS validated_revenue

FROM northwind_db.order_details od

JOIN northwind_db.orders o ON od.orderID = o.orderID;
```

- **Audit Result:** 2,155 Line Items | $1,265,793.04 Validated Revenue.
- **Analyst Note:** This query confirmed the $1.27M baseline and ensured that the row-level 15%-25% discounts were mathematically sound before modelling began in Power BI. Without discount, the revenue will be $1.35M as a gross revenue.